Transfer functions

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1 Introduction

- In this part we'll look in more detail at transfer functions
- We start by examining the definition of *"system,"* as an element that takes an input signal and produces an output signal.
- Pole placement design (our first design method!)
- Mathematical tools for complex numbers
- Rational Polynomials (transfer functions and Laplace transforms) have three standard forms,

  1. Rational Polynomial form (what we find when we derive $G_{uy}(s)$)
  2. Sum-of-terms form (needed for partial fraction expansion)
  3. Product-of-terms form (used by many analysis tools).

2 What is a System

- A ***system*** is something that maps one (or more) input signals to one (or more) output signals, for example, as seen in figure 1, an aircraft maps inputs to outputs.

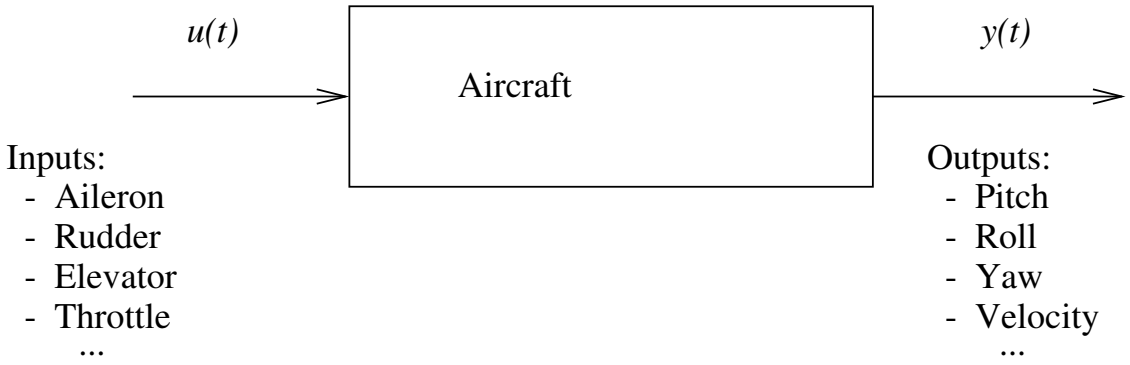
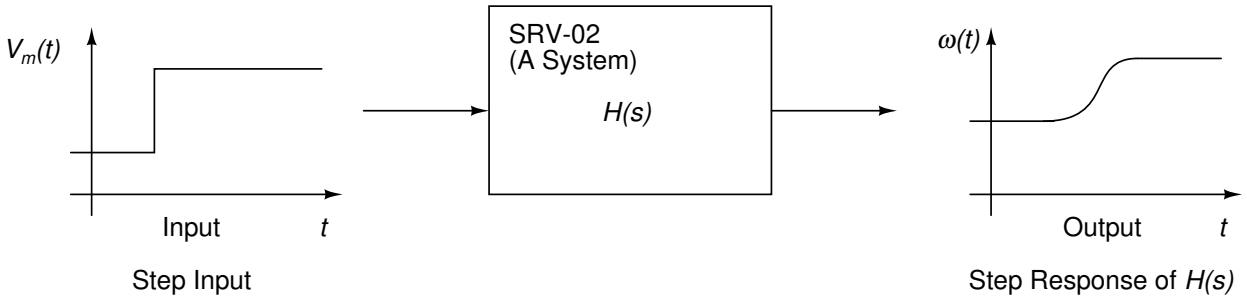


Figure 1: An example system. A system converts input signals to output signals.

- We can also consider the **system response**:
  - The output for a given input



2.1 General approaches to controls analysis and design

- There are two general approaches controls analysis and design.

Mathematical Approach	Simulation or Experimental Approach
* Approximate	* Precise
* General results	* Results limited to system studied
- Consider all possible inputs	- Consider one specific input at a time
* Design tools and insight	* Little design insight

- There are three domains for mathematical analysis.

Mathematical Analysis Domains	Course	Uses Transfer Function
1. S-Plane (poles, zeros)	EE 474	$\surd$
2. Frequency Response (Bode plot)	EE 574	$\surd$
3. State Space Analysis	EE 574	

- In EE 474 we are most concerned with analysis tools and design methods; and so we focus on mathematical analysis.

  Linear Time-Invariant systems (the kind considered in EE 474) can be described in two ways: LTI

  1. By one or several Linear Constant Coefficient Differential Equations (LCCDE), for example, of the systems considered in chapter 2.

  2. A Transfer Function (TF)

3 Transfer functions (Franklin *et al.* section 3.2 (7th ed.))

- Every system described by an Linear Constant Coefficient Differential Equation (LCCDE) has a transfer function, and vice versa.

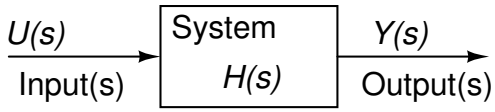


Figure 2: System with transfer function $H(s)$.

- The T.F. is derived as the ratio of Laplace Transforms: Definition of TF

  For any physical signals and system, both the input signal and output signal have a Laplace Transform. Referring to figure 2:

$$U(s) = \mathcal{L}\{u(t)\} \qquad Y(s) = \mathcal{L}\{y(t)\}$$

  The transfer function is the ratio of the output and input Laplace Transforms:

$$H(s) = \frac{Y(s)}{U(s)} \quad ; \quad \frac{\text{LT of Output}}{\text{LT of Input}} \quad ; \quad \left[\frac{\text{Units of Output}}{\text{Units of Input}}\right] \tag{1}$$

- The transfer function has units of output over input, or *Gain*. Units of a TF

3.1 Determining the transfer Function

- The Transfer Function is the ratio of the Laplace Transform of the output signal over the Laplace Transform of the input signal.
  - The TF is gain in the frequency domain, or the ***Generalized Gain***.
- We can form the TF directly from the differential equation.

3.1.1 Example: determining the transfer function

Given a system governed by:

$$2\ddot{y}(t)+3\dot{y}(t)+4y(t)=5\dot{u}(t)+6u(t) \tag{2}$$

Taking the L.T. of both sides gives:

$$\mathcal{L}\{2\ddot{y}(t)+3\dot{y}(t)+4y(t)\}=\mathcal{L}\{5\dot{u}(t)+6u(t)\} \tag{3}$$

Using linearity of the L.T. and derivative theorem:

$$2s^2Y(s)+3sY(s)+4Y(s) \;=\; 5sU(s)+6U(s) \tag{4}$$

Algebra: (factoring out $Y(s)$ and $U(s)$)

$$Y(s)\left(2s^2+3s+4\right) \;=\; U(s)\left(5s+6\right) \tag{5}$$

Algebra (writing output over input):

$$\frac{Y(s)}{U(s)}=G_{uy}(s)=\frac{5s+6}{2s^2+3s+4} \tag{6}$$

!! That's all there is to forming the transfer function !!

Now, if we know the TF of a system, and know the LT of the input, the LT of the output is given by:

$$Y(s)=G_{uy}(s)\,U(s) \tag{7}$$

3.1.2 Example, deriving the transfer function of a drive

The model and transfer functions

- In this section, the transfer function of a series feedback system is derived. The system is illustrated in figure 3.

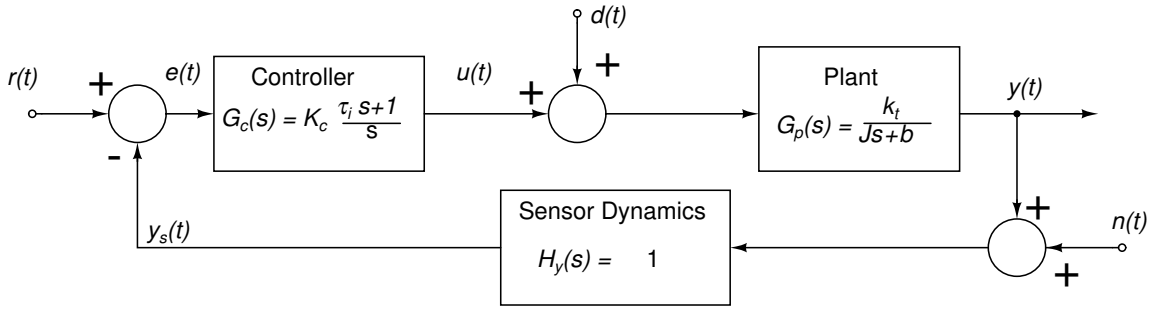


Figure 3: Series feedback system.

- Tables of signals and parameters are given

Symbol	Description	Units		Symbol	Description	Units
$r(t)$	Reference input	$\left[\frac{\text{radians}}{\text{second}}\right]$		$y(t)$	Plant output	$\left[\frac{\text{radians}}{\text{second}}\right]$
$e(t)$	Error	$\left[\frac{\text{radians}}{\text{second}}\right]$		$y_s(t)$	Sensed plant output	$\left[\frac{\text{radians}}{\text{second}}\right]$
$u(t)$	Control signal	[Amps]		$n(t)$	Sensor noise	$\left[\frac{\text{radians}}{\text{second}}\right]$
$d(t)$	Disturbance	[Amps-equivalent]				

Table 1: Signals

Symbol	Description	Units		Symbol	Description	Units
K_c	Controller gain	$\left[\frac{\text{Amps}}{\text{radian}}\right]$		k_t	Motor constant	$\left[\frac{\text{N-m}}{\text{Amp}}\right]$
τ_i	Integrator time constant	[s]		J	Inertia	$\left[\frac{\text{N-m-}s^2}{\text{radian}}\right]$
				b	damping	$\left[\frac{\text{N-m-s}}{\text{radian}}\right]$

Table 2: Parameters

- The units of each transfer function can be found in two ways:

  1. From the input and output signals

$$G_c(s) \quad : \quad \left[\frac{\text{amps}}{\text{radian/second}}\right], \tag{8}$$

  2. By considering the parameters of the equation for the transfer function

$$\begin{aligned} G_c(s) = K_c\frac{\tau_i s+1}{s} \quad &: \quad \left[\frac{\text{amps}}{\text{radian}}\right] \times \left[\frac{s\,s^{-1}+[\cdot]}{s^{-1}}\right] \\ &= \left[\frac{\text{amp-sec}}{\text{radian}}\right] = \left[\frac{\text{amp}}{\text{radian/sec}}\right] \end{aligned} \tag{9}$$

     Notice that 1 is dimensionless, and so $\tau_i s$ must be also.

- For the plant

  1.

$$G_p(s) \quad : \quad \left[\frac{\text{radians/second}}{\text{amp}}\right] \tag{10}$$

  2.

$$\begin{aligned} G_p(s) = \frac{k_t}{Js+b} \quad &: \quad \left[\frac{\text{N-m/amp}}{\frac{\text{n-m-}s^2}{\text{radian}}s^{-1}+\frac{\text{n-m-s}}{\text{radian}}}\right] = \frac{\text{N-m}}{\text{N-m}}\,\frac{1}{\text{amps}}\,\frac{1}{\text{sec/radian}} \\ &= \left[\frac{\text{radian/second}}{\text{amp}}\right] \end{aligned}$$

Closed-loop transfer functions There are many possible closed-loop transfer functions:

- The system may have several inputs
- Each signal in the system can be considered as an output

Notice that every input affects every output.

Several possible closed-loop transfer functions are

$$T_{ry}(s) = \frac{Y(s)}{R(s)} \quad \text{System response}$$

$$T_{re}(s) = \frac{E(s)}{R(s)} \quad \text{Error response}$$

$$T_{dy}(s) = \frac{Y(s)}{D(s)} \quad \text{Disturbance response}$$

For each of these, the closed loop transfer function is give by

$$\frac{\text{Forward path gain}}{1 + \text{Loop Gain}}$$

$$T_{ry}(s) = \frac{\text{Gain from } r(t) \text{ to } y(t)}{1+\text{Loop Gain}} = \frac{G_c G_p}{1 + G_c G_p H_y} \tag{11}$$

$$T_{re}(s) = \frac{\text{Gain from } r(t) \text{ to } e(t)}{1+\text{Loop Gain}} = \frac{+1}{1 + G_c G_p H_y} \tag{12}$$

$$T_{dy}(s) = \frac{\text{Gain from } d(t) \text{ to } y(t)}{1+\text{Loop Gain}} = \frac{G_p}{1 + G_c G_p H_y} \tag{13}$$

Checking units of a transfer function, before algebraically reducing it to rational polynomial form We can check units of the transfer function in the form of Eqn (11), for example,

$$T_{ry}(s) = \frac{G_c G_p}{1 + G_c G_p H_y} \tag{14}$$

1. From looking at the input and output signals,

$$T_{ry}(s) \; : \; \left[\frac{\text{rad/sec}}{\text{rad/sec}}\right] = [\cdot]$$

   The input-to-output transfer function is dimensionless.

2. By combining the units of individual terms:

$$\begin{aligned} T_{ry}(s) &= \frac{G_c G_p}{1 + G_c G_p H_y} \; : \; \left[\frac{\left[\frac{\text{amps}}{\text{rad/sec}}\right]\left[\frac{\text{rad/sec}}{\text{amp}}\right]}{1 + \left[\frac{\text{amps}}{\text{rad/sec}}\right]\left[\frac{\text{rad/sec}}{\text{amp}}\right]\left[\frac{\text{rad/sec}}{\text{rad/sec}}\right]}\right] \\ &= \frac{[\cdot]}{1 + [\cdot]} = [\cdot] \end{aligned}$$

Putting the transfer function in rational-polynomial form In equation (14), each of the individual TFs ($G_c(s)$, $G_p(s)$, etc.) within the overall TF ($T_{ry}(s)$) is a rational polynomial. So when we plug them in, we get fractions within the fraction

$$T_{ry}(s) = \frac{G_c G_p}{1 + G_c G_p H_y} = \frac{K_c \frac{\tau_i s+1}{s} \frac{k_t}{Js+b}}{1 + K_c \frac{\tau_i s+1}{s} \frac{k_t}{Js+b}} \tag{15}$$

By multiplying

$$T_{ry}(s) \times \frac{s(Js+b)}{s(Js+b)} = \frac{K_c \frac{\tau_i s+1}{s} \frac{k_t}{Js+b}}{1 + K_c \frac{\tau_i s+1}{s} \frac{k_t}{Js+b}} \frac{s(Js+b)}{s(Js+b)}$$

we clear the fractions within the fraction, to get

$$T_{ry}(s) = \frac{K_c(\tau_i s+1) k_t}{s(Js+b) + K_c(\tau_i s+1) k_t} \tag{16}$$

Equation (16) is not in rational polynomial form. For example, we are not ready to find the roots of the denominator polynomial (called the ***poles***). Multiplying out and combining the expressions, gives

$$T_{ry}(s) = \frac{K_c k_t(\tau_i s+1)}{Js^2 + (b + K_c k_t \tau_i) s + K_c k_t} \tag{17}$$

Now we may check units. This is the most important point at which to do so, since it is the end of the derivation of the closed-loop transfer function.

$$T_{ry}(s) : \frac{\left[\frac{\text{amps}}{\text{radian}}\right]\left[\frac{\text{N-m}}{\text{amp}}\right]\left([s]\left[s^{-1}\right]+[\cdot]\right)}{\left[\frac{\text{n-m-}s^2}{\text{radian}}\right] s^{-2} + \left(\left[\frac{\text{n-m-s}}{\text{radian}}\right] + \left[\frac{\text{amps}}{\text{radian}}\right]\left[\frac{\text{N-m}}{\text{amp}}\right][\text{s}]\right)[\text{s}^{-1}] + \left[\frac{\text{amps}}{\text{radian}}\right]\left[\frac{\text{N-m}}{\text{amp}}\right]} \tag{18}$$

The term $k_c k_t$ appears in three places. This term has units

$$k_c k_t : \left[\frac{\text{amps}}{\text{radian}}\right]\left[\frac{\text{N-m}}{\text{amp}}\right] = \left[\frac{\text{N-m}}{\text{radian}}\right]$$

Using this and combining terms with equal units (the only kind that can be added !), gives

$$T_{ry}(s) : \frac{\left[\frac{\text{n-m}}{\text{radian}}\right]}{\left[\frac{\text{n-m}}{\text{radian}}\right] + \left[\frac{\text{n-m}}{\text{radian}}\right] + \left[\frac{\text{n-m}}{\text{radian}}\right]} = [\cdot] \tag{19}$$

and the $T_{ry}(s)$ is dimensionless, as it should be.

The most interesting term, dimensionally, is the middle one in the denominator:

$$(b + K_c\, k_t\, \tau_i)\, s \tag{20}$$

It is not obvious that units will balance in this term, which combines mechanical, electrical and control parameters. However, we see from the expansion of the units

$$\left(\left[\frac{\text{n-m-s}}{\text{radian}}\right] + \left[\frac{\text{amps}}{\text{radian}}\right] \left[\frac{\text{N-m}}{\text{amp}}\right] [\text{s}] \right) \left[\text{s}^{-1}\right] = \left[\frac{\text{n-m}}{\text{radian}}\right] \tag{21}$$

that all is as it should be.

4 Pole-placement design

Controller Design Method

- Now that we can build closed-loop transfer functions, it is time to introduce one simple controller design method: pole-placement design.

- If we have N_c controller parameters, we can, in general, set the N_c controller parameters to place N_c closed-loop poles at desired locations on the S-plane.

- Pole-placement design is most straightforward when the number of parameters equals the number of closed-loop system poles.

  1. Determine the equation for the closed-loop transfer function;

     (a) Divide through by a_0, so the characteristic equation is monic.

  2. Choose the desired pole locations.

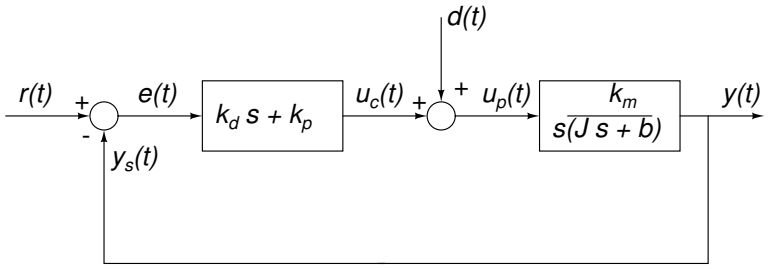
  3. Based on the desired pole locations, determine the monic characteristic equation.

  4. Match up the polynomial coefficients, and solve for the controller parameters.

- Pole placement design used in lab part 5.

4.1 Example: Pole-placement design

- Consider the example of figure 4.



Param	Value	Units
k_m	2.5	[rad/volt]
J	2.0	[N-m-s^2/rad]
b	1.5	[N-m-s/rad]
k_p	?	[volts/rad]
k_d	?	[volts/rad/sec]

Figure 4: Proportional-Derivative (PD) control example.

1. Form the transfer function (denominator monic) and check units

$$T_{ry}(s) = \frac{k_d\,k_m\,s + k_p\,k_m}{J\,s^2 + (b + k_d\,k_m)\,s + k_p\,k_m} = \frac{k_d\,k_m/J\,s + k_p\,k_m/J}{s^2 + (b/J + k_d\,k_m/J)\,s + k_p\,k_m/J} \tag{22}$$

2. Determine the desired pole locations: $p_{1,2} = -3 \pm j4$

3. Form desired monic characteristic polynomial:

$$\Delta(s) = (s - p_1)(s - p_2) = s^2 + 6s + 25 \tag{23}$$

4. Set the system char. Eqn, (22), equal to the desired char. Eqn, (23):

$$s^2 + (b/J + k_d\,k_m/J)\,s + k_p\,k_m/J \;=\; s^2 + 6s + 25 \tag{24}$$

For Eqn (24) to be true for all s, it must balance for each power of s:

$$\begin{aligned} s^2: \quad & 1 &&= 1 \\ s^1: \quad & k_p\,k_m/J &&= 25 \\ s^0: \quad & b/J + k_d\,k_m/J &&= 6 \end{aligned}$$

Giving the controller parameters:

$$\begin{aligned} k_p &= 25\,\frac{J}{k_m} = 20 \text{ [volts/rad]} \\ k_d &= \left(6 - \frac{b}{J}\right)\frac{J}{k_m} = 4.2 \text{ [volt-sec/rad]} \end{aligned}$$

5 Properties of the transfer function

For something so simple as going from Eqn (2) to Eqn (6), the transfer function has an amazing wealth of properties.

We list properties here. These properties are worked out fully in subsequent lectures.

1. Components of a Transfer Function

   (a) All information about the signals is included in the Laplace Transforms $U(s)$ and $Y(s)$;

   (a) **All information about the system is included in the transfer function $T_{ry}(s)$!** Key TF Fact

   $\Rightarrow$ when we evaluate properties the transfer function, we are evaluating the response of the system, **without** reference to any specific input or output signal.

   $\Rightarrow$ we can study $T_{ry}(s)$ and design a controller without ever knowing $U(s)$ of $Y(s)$. (The controls engineer almost never knows the actual signals $u(t)$ or $y(t)$).

2. The transfer function has units of gain. For example, for a voltage-driven, velocity-controlled motor servo, the TF has units:

$$G_p(s): \qquad \left[\frac{\text{rad/sec}}{volt}\right]$$

3. A transfer function is a rational polynomial, for example: TF is Rat Poly

$$T_{ry}(s) = \frac{5s+6}{2s^2+3s+4} = \frac{\text{Numerator Polynomial in } s}{\text{Denominator Polynomial in } s} \qquad (25)$$

4. A TF is neither more nor less than a function of s. Any algebra we can do with a function of s, we can do with a TF.

Properties of the Transfer Function (continued)

5. A TF can be written with a single symbol, such as $G_p(s)$ or $T_{ry}(s)$, or, to show that a TF is from the u input to the y output, we can represent the TF by:

$$G_{uy}(s) = \frac{Y(s)}{U(s)}$$

EE474 notation:

$$\begin{aligned} T_{ry}(s) &: \text{closed-loop transfer function} \\ G_{uy}(s) &: \text{open-loop blocks} \end{aligned}$$

6. Transfer functions depend on linearity; non-linear systems **do not have** a transfer function.

5.1 Transfer functions, superposition and Multi-Input Multi-Output (MIMO) systems

- Many systems are either multi-input, multi-output, or both (MIMO).
- Terminology
  - SISO: Single-Input Single-Output (what we mainly consider in EE 474, and a majority of control loops in service)
  - MIMO: Multi-Input Multi-Output
- Consider figure 5 as an automobile cruise control.

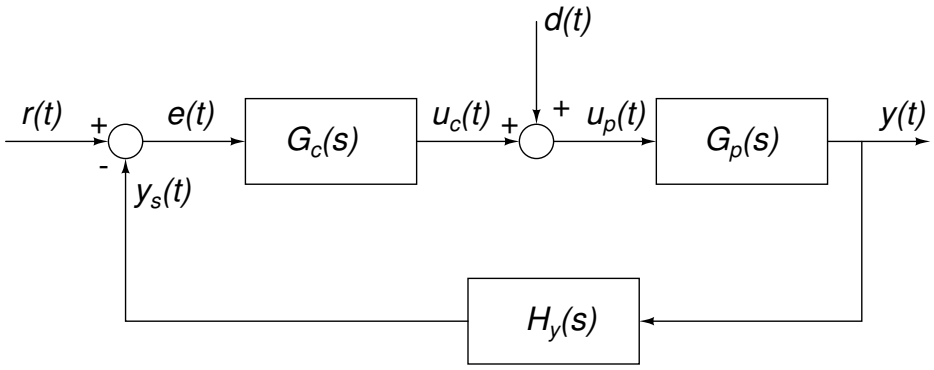


Figure 5: System with two inputs and three outputs has six transfer functions. Any signal can be considered an output.

  - Signals $r(t)$ and $d(t)$ are the reference and disturbance inputs.
  - Any signal (such as $e(t)$, $u_c(t)$ or $y(t)$) can be considered an output (or consider 3 outputs at once).

– MIMO systems are very simply represented with State-Space models. For example, using figure 6 with

$$\mathbf{u}(t) = \begin{bmatrix} r(t) \\ d(t) \end{bmatrix} \qquad \mathbf{y}(t) = \begin{bmatrix} e(t) \\ u_c(t) \\ y(t) \end{bmatrix}$$

a 2-input, 3-output system is easily represented.

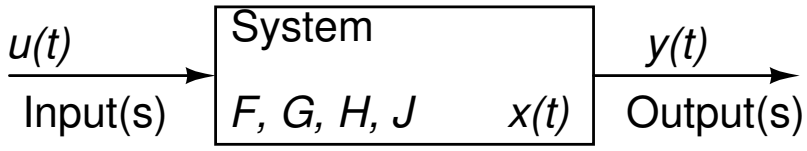


Figure 6: Same system in State Variable form (EE/ME 574).

- Using transfer functions for a MIMO system, we need to consider a transfer function between every input and every output, as shown in figure 7.

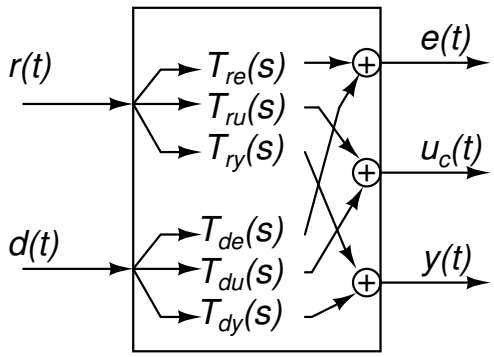


Figure 7: System with two inputs and three outputs has six transfer functions.

1. The multiple transfer functions work together by superposition:

$$E(s) = T_{re}(s)\,R(s) + T_{de}(s)\,D(s) \tag{26}$$
$$U_c(s) = T_{ru}(s)\,R(s) + T_{du}(s)\,D(s) \tag{27}$$
$$Y(s) = T_{ry}(s)\,R(s) + T_{dy}(s)\,D(s) \tag{28}$$

2. Individual TFs are found using superposition:

$$T_{ry}(s) = \frac{Y(s)}{R(s)} \qquad (\text{assuming } d(t) = 0)$$

$$T_{dy}(s) = \frac{Y(s)}{D(s)} \qquad (\text{assuming } r(t) = 0)$$

(see example below).

Note: $T_{ry}(s)$ is the TF from $r(t)$ to $y(t)$. It is customary to write this TF as $Y(s)/R(s)$. But when we put this notation in Eqn (28) we get: Important Note

$$Y(s) = \frac{Y(s)}{R(s)} R(s) + \frac{Y(s)}{D(s)} D(s) \tag{29}$$

which appears to (erroneously) say $Y(s) = 2Y(s)$.

Remember: $Y(s)/R(s)$ is just a symbol for a TF, $T_{ry}(s)$ in this case. Using the notation of Eqn (29) inappropriately suggests $Y(s) = 2Y(s)$ (the superposition may be obscured). If we write

$$Y(s) = T_{ry}(s)\, R(s) + T_{dy}(s)\, D(s) \tag{30}$$

the superposition is more clear.

5.2 What does a transfer function tell us about the response of a system and the performance of a closed-loop system

- For any input:
  - System order
  - Stability
- For a Sinusoidal Input:
  - Magnitude and phase of the sinusoidal output, given a sinusoidal input (the Bode plot)
- For a Step Input:
  - Steady State Error
  - Damping
  - Rise Time
  - Settling Time
- Pole-zero constellation

6 Some needed math tools

6.1 Rational Polynomials

- A rational polynomial is a function which is a "ratio" of polynomials, such as

$$H(s) = \frac{2s^2 + 10s + 12}{s^2 + 12s + 35} \tag{31}$$

- Both Laplace transforms and transfer functions are rational polynomials in s !

6.2 Poles and Zeros

- Rational polynomials have numerator and denominator polynomials.
- The **roots** of the numerator and denominator polynomials will play special roles, they are called:

  Poles: the roots of the denominator polynomial.

  Zeros: the roots of the numerator polynomial.

  For example, given:

$$H(s) = \frac{2s^2 + 10s + 12}{s^2 + 12s + 35} \tag{32}$$

- Find the zeros by solving:

$$2s^2 + 10s + 12 = 0$$

  Which gives:

$$z_1 = -2\,,\ \ z_2 = -3$$

- Find the poles by solving:

$$s^2 + 12s + 35 = 0$$

  Which gives:

$$p_1 = -5\,,\ \ p_2 = -7$$

- Any polynomial may be factored in terms of its roots :

$$2s^2+10s+12 = 2\left(s^2+5s+6\right) = 2\left(s+2\right)\left(s+3\right)$$

$$s^2+12s+35 = \left(s+5\right)\left(s+7\right)$$

- We can also write the TF in terms of the poles and zeros, giving product-of-terms form:

$$\begin{aligned} H(s) &= = \frac{2\left(s+2\right)\left(s+3\right)}{\left(s+5\right)\left(s+7\right)} = 2\frac{\left(s-z_1\right)\left(s-z_2\right)}{\left(s-p_1\right)\left(s-p_2\right)} \\ &= 2\frac{\left(s-(-2)\right)\left(s-(-3)\right)}{\left(s-(-5)\right)\left(s-(-7)\right)} \end{aligned} \tag{33}$$

where the last expression illustrates why the quantities are positive in the TF, but we subtract the p_i and z_i in the product-of-terms form.

6.2.1 The Poles and Zeros may be either real or complex pairs

- Both the numerator and denominator above had real roots. But the roots of polynomials may be complex. **Complex roots brings complex arithmetic into controls analysis.** Complex Numbers Enter In

  For example, given:

$$G(s) = \frac{1}{s^2 + 2s + 10} \tag{34}$$

- The poles are given by the solutions of the characteristic equation:

$$s^2 + 2s + 10 = 0$$

  Applying the quadratic equation gives:

$$p_i = \frac{-2 \pm \sqrt{2^2 - 4 * 10}}{2} = \frac{-2 \pm \sqrt{-36}}{2} = -1 \pm \sqrt{-1}\sqrt{9} = -1 \pm j3 \tag{35}$$

  where $j = \sqrt{-1}$ is the ***imaginary*** number. Eqn (35) gives:

$$\begin{aligned} p_1 &= -1 + 3j \\ p_2 &= -1 - 3j \end{aligned}$$

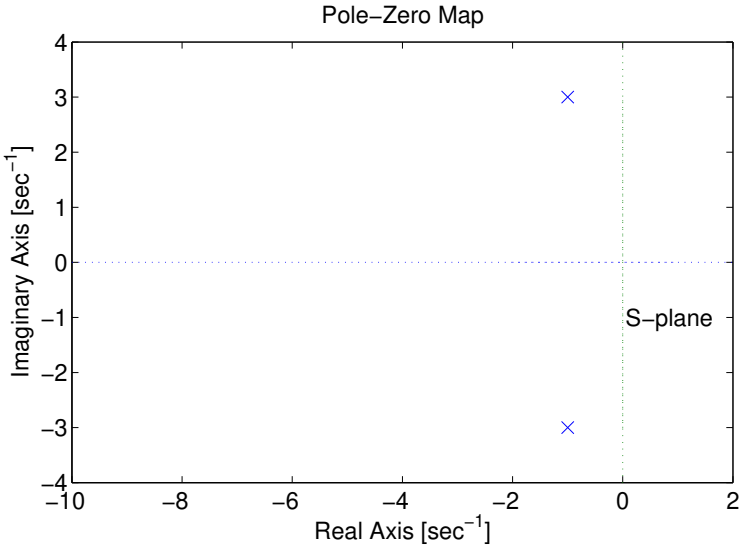


Figure 8: Pole-zero map for $G(s)$ in Eqn (34).

6.3 Quick review of Complex Numbers

- Because the poles, zeros and residues of Laplace transforms and transfer functions can (and often will) be complex, complex numbers play a great role controls analysis.

- Complex numbers have real and imaginary parts, such as:

$$A = -3+4\,j \qquad\qquad B = 5-3\,j$$

6.3.1 Arithmetic:

- Operations $+$, $-$, $\times$ and $\div$ are defined as for real numbers, but carry j as an unknown:

$$\begin{aligned}
C = A+B &= (-3+4\,j)+(5-3\,j) = (-3+5)+j\,(4-3) = 2+1\,j \\
D = A-B &= (-3+4\,j)-(5-3\,j) = (-3-5)+j\,(4+3) = -8+7\,j \\
E = A\times B &= (-3+j4)\times(5-j3) = -15+9\,j+20\,j-12\,j^2 = -3+29\,j \\
F = A/B &= \frac{-3+4\,j}{5-3\,j} = \frac{(-3+4\,j)\,(5+3\,j)}{(5-3\,j)\,(5+3\,j)} = \frac{-27+11\,j}{34} = -\frac{27}{34}+\frac{11}{34}\,j
\end{aligned}$$

  where $j^2 = -1$ was used in the multiplication and division.

- Be sure you can do basic operations with complex numbers.

  - Calculators do complex arithmetic (with specific numbers), **but**

  - We'll be developing tools and doing analysis with complex numbers; which is to say, algebra with complex numbers.

6.3.2 Rectangular and polar coordinates:

- A complex number represents a point on the complex plane, as illustrated in figure 9. This point can be represented in two forms:

Representation	Two parts	Example
Rectangular	real, imaginary	$A = a + jb$
Polar	magnitude, angle	$A = M\angle\theta$

The conversion back and forth is:

rect. -> polar	$M = \sqrt{a^2 + b^2}$;	$\theta = \text{atan2}(b, a)$
polar -> rect.	$a = M\cos(\theta)$;	$b = M\sin(\theta)$

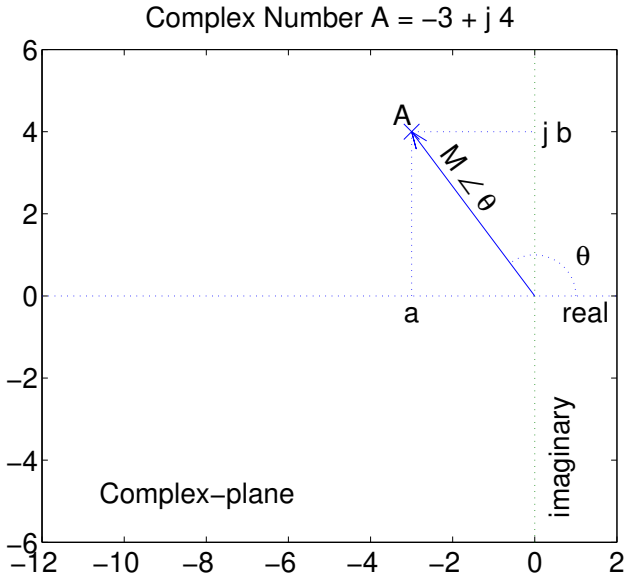
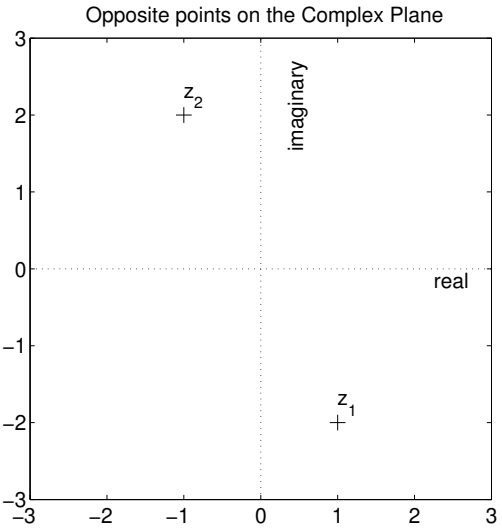


Figure 9: A complex number in rectangular and polar coordinates

6.3.3 Polar to Rectangular conversion and the atan2() function

- Notice the `atan2()` function in the conversion from rectangular to polar coordinates.
- Consider figure 10. Points z_1 and z_2 do not represent the same number:



$$z_1 = 1-2j = 2.2\angle -63^o$$
$$z_2 = -1+2j = 2.2\angle 117^o$$

Figure 10: Two opposite points in the complex plane.

- Now consider the arc-tangent function:

$$\begin{aligned} \text{range of atan}() : &\quad -90^o \cdots +90^o \\ z_1 : &\quad \text{atan}\left(\frac{-2}{1}\right) = \text{atan}(-2) = -63^o \\ z_2 : &\quad \text{atan}\left(\frac{2}{-1}\right) = \text{atan}(-2) = -63^o \end{aligned}$$

  With the `atan()` function, the angles of both z_1 and z_2 would come out to be -63°, which would make the two numbers the same in polar coordinates !

- The atan2() function is also called the 'four-quadrant arc-tangent' function; and must be used for rectangular to polar conversion.

$$\begin{aligned} \text{range of atan2}() : &\quad -180^o \cdots +180^o \\ z_1 : &\quad \text{atan2}(-2, 1) = -63^o \\ z_2 : &\quad \text{atan2}(2, -1) = +117^o \end{aligned}$$

6.3.4 Complex arithmetic in polar coordinates

- Operations $\times$ and $\div$ can also be done with complex numbers in polar form. In the general case, given two complex number G and H, the product and ratio are given by:

$$N = G \times H = (M_G \angle \theta_G)(M_H \angle \theta_H) = M_G M_H \angle (\theta_G + \theta_H) \quad (36)$$

$$P = G \div H = \frac{(M_G \angle \theta_G)}{(M_H \angle \theta_H)} = \frac{M_G}{M_H} \angle (\theta_G - \theta_H) \quad (37)$$

  So multiplication is

  - multiplication of the magnitudes and addition of the angles;

  Division is

  - division of the magnitudes and subtraction of the angles.

- For example:

$$G = 5 \angle 20^o \qquad H = 7 \angle -45^o$$

  then

$$N = G \times H = (5 \angle 20^o)(7 \angle -45^o) = 35 \angle -25^o$$
$$P = G \div H = \frac{(5 \angle 20^o)}{(7 \angle -45^o)} = \frac{5}{7} \angle 65^o$$

6.3.5 Euler's formula

An important relation between complex numbers comes out of the series forms for cos, sin and the exponential functions:

$$\cos(x) = 1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 - \cdots \tag{38}$$

$$\sin(x) = x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \cdots \tag{39}$$

$$\exp(x) = 1 + x + \frac{1}{2!}x^2 + \frac{1}{3!}x^3 + \frac{1}{4!}x^4 + \frac{1}{5!}x^5 + \cdots \tag{40}$$

Consider Eqn (40) if x is imaginary:

$$\begin{aligned}\exp(jx) &= 1 + (jx) + \frac{1}{2!}(jx)^2 + \frac{1}{3!}(jx)^3 + \frac{1}{4!}(jx)^4 + \frac{1}{5!}(jx)^5 + \cdots \\ &= 1 + jx - \frac{1}{2!}x^2 - \frac{1}{3!}jx^3 + \frac{1}{4!}x^4 + \frac{1}{5!}jx^5 + \cdots\end{aligned}$$

Using

$$j^2 = -1\,, \quad j^3 = -j\,, \quad j^4 = 1\,,\ j^5 = j\,,\ j^6 = -1\,, \quad \ldots \tag{41}$$

Grouping terms into real and imaginary terms gives:

$$\begin{aligned}\exp(jx) &= 1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 - \cdots + \quad j\left(x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \cdots\right) \\ &= \cos(x) + j\sin(x)\end{aligned} \tag{42}$$

where the last identity is known as Euler's formula .

- Using Euler's formula we can also derive the relations:

$$\sin(\omega t) = \frac{1}{2j}\left(e^{j\omega t} - e^{-j\omega t}\right) \qquad \cos(\omega t) = \frac{1}{2}\left(e^{j\omega t} + e^{-j\omega t}\right)$$

6.4 Complex conjugates

- If $A = a + jb$, then the complex conjugate, written A^*, is given by $A^* = a - jb$. For example:

Rectangular:	$A = -3 + j4$	$A^* = -3 - j4$
Polar:	$A = 5\angle 126.9^o$	$A^* = 5\angle -126.9^o$

- Notice that for

$$G(s) = \frac{1}{s^2 + 2s + 10} \tag{43}$$

The poles are:

$$\begin{aligned} p_1 &= -1 + 3j \\ p_2 &= -1 - 3j \end{aligned}$$

which is a complex conjugate pair.

- If a polynomial has real coefficients, its complex roots (if any) **must** come in complex conjugate pairs.
  - Differential equations that describe physical systems have real coefficients.

$\Rightarrow$ Poles and Zeros come in complex conjugate pairs, if a system has a pole

$$p_1 = -5 + 7j$$

it must also have a pole:

$$p_2 = -5 - 7j$$

- The consequence of all complex terms coming in complex conjugate pairs is that when the terms are re-combined, the imaginary parts will always appear in equal and and opposite parts and cancel out.
- This must happen because all physical signals are real.

7 A TF is simply a function of s

- Any algebra we can do with a function of s, we can do with a TF.
  - We can add transfer functions: equivalent to system blocks in parallel.
  - We can multiply transfer functions: equivalent to system blocks in series.
  - We can re-arrange terms. For example, consider what happens when a system has:
    * A zero, which moves further to the left, while
    * The DC gain remains constant (see pz-maps in figure 11)

$$H_1(s) = \frac{s+5}{s^2+6s+25} \qquad H_2(s) = \frac{1}{3}\,\frac{s+15}{s^2+6s+25} \tag{44}$$

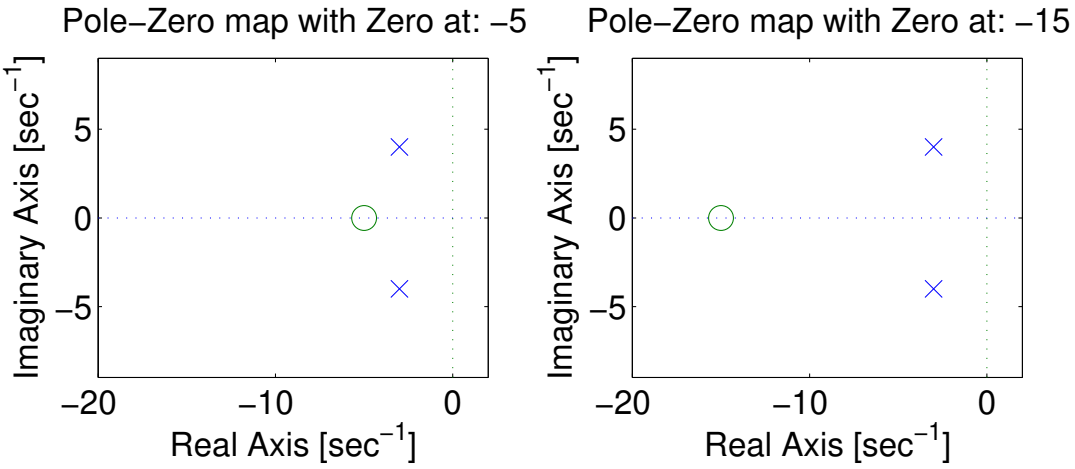


Figure 11: A system with a shifted zero, K_{DC} remains unchanged.

  - Rather than bringing out the extra $1/3$ term, we could write:

Algebra on TF Example

$$H_1(s) = 5\,\frac{(s/5+1)}{s^2+6s+25} \qquad H_2(s) = 5\,\frac{(s/15+1)}{s^2+6s+25} \tag{45}$$

  - Which helps make clear that only the zero moves; the DC gain doesn't change.

What are K_{Bode} and K_{rlg} for $H_1(s)$ and $H_2(2)$?

Student Exercise

8 Rational Polynomials (TFs and LTs) have three standard forms

Standard forms for either $H(s)$ or $Y(s)$

There are three equivalent standard forms for expressing a TF or LT, each having important uses.

1. Rational polynomial form
2. Sum-of-terms form
3. Product-of-terms form

As an example TF, consider:

$$H(s) = \frac{35s^2 + 82s + 48}{18s^3 + 47s^2 + 66s + 40} \tag{46}$$

1. Rational polynomial form

$$H(s) = \frac{35s^2 + 82s + 48}{18s^3 + 47s^2 + 66s + 40} = \frac{1.94s^2 + 4.56s + 2.67}{s^3 + 2.61s^2 + 3.67s + 2.22} \tag{47}$$

2. Sum-of-terms form

$$H(s) = \frac{0.0035}{s + 1.11} + \frac{1.94s + 2.39}{s^2 + 1.5s + 2} \tag{48}$$

3. Product of terms form

$$H(s) = \frac{(7s + 8)(5s + 6)}{(9s + 10)(2s^2 + 3s + 4)} \tag{49}$$

8.1 Rational Polynomial Form

- "Rational Polynomial" means the expression is the ratio of two polynomials.
- Rational Polynomial form ("Rat-Poly" for short) is the standard form for Transfer Functions, and comes directly from a differential equation.

8.1.1 General rational polynomial form:

$$H(s) = \frac{b_m s^m + \cdots + b_1 s + b_0}{a_n s^n + a_{n-1} s^{n-1} + \cdots + a_1 s + a_0} \tag{50}$$

8.1.2 Rational polynomial form example (with $a_n \neq 1$):

$$H(s) = \frac{35 s^2 + 82 s + 48}{18 s^3 + 47 s^2 + 66 s + 40} \tag{51}$$

8.1.3 Rational polynomials can be put in root locus or Bode form

- As we've seen, rational polynomials can be setup 2 ways, depending on how the constant is brought out:

Root-locus form: $$H(s) = K_{rlg} H_1(s) = \frac{35}{18} \frac{1 s^2 + \frac{82}{35} s + \frac{48}{35}}{1 s^3 + \frac{47}{18} s^2 + \frac{66}{18} s + \frac{40}{18}}$$

Bode form: $$H(s) = K_{Bode} \frac{1}{s^{N_o}} H_2(s) = \frac{48}{40} \frac{1}{s^0} \frac{\frac{35}{48} s^2 + \frac{82}{48} s + 1}{\frac{18}{40} s^3 + \frac{47}{40} s^2 + \frac{66}{40} s + 1}$$

so $K_{rlg} = \frac{35}{18}$ while $K_{Bode} = \frac{48}{40}$.

- In root-locus form, the leading coefficients (highest powers of s) are 1.
- In Bode form, the lowest order non-zero coefficients are 1.

8.2 Sum-of-terms form (or partial fraction expansion form, or parallel-realization form)

- Sum-of-terms form breaks the TF (or LT) down into its simplest individual elements.
- This form is used to find $y(t)$, since each of the individual terms can be looked up in tables.
- Sum-of-terms form requires finding the residues, C_1, C_2, etc.
- A high-order filter with TF $H(s)$ can be built by building a low-order filter for each term, and combining them parallel.

8.2.1 General sum-of-terms form :

$$H(s) = \frac{C_1}{s-p_1} + \frac{C_2}{s-p_2} + \frac{C_3}{s-p_3} \tag{52}$$

Or

$$H(s) = \frac{C_1}{s-p_1} + \frac{C_2 s + C_3}{s^2 + a_1 s + a_2} \tag{53}$$

8.2.2 Sum-of-terms form example:

Three steps to determine the sum-of-terms form are:

1. Find the poles:

$$\text{roots}\left(\left[18\,s^3+47\,s^2+66\,s+40\right]\right) = \left[\begin{array}{cc} -1.11 & -0.75 \pm 1.20\,j \end{array}\right]$$

2. Write $H(s)$ in the form of Eqn (52) (with complex-conjugate poles p_2 , p_2^*):

$$\begin{aligned} H(s) &= \frac{C_1}{s+1.11} + \frac{C_2\,s+C_3}{s^2+1.5\,s+2} \\ &= \frac{C_1}{s+1.11} + \frac{C_a}{s+0.75-1.20\,j} + \frac{C_a^*}{s+0.75+1.20\,j} \end{aligned} \tag{54}$$

   where the relationship between C_a and C_2 and C_3 is a topic of Part 3c.

   - The denominator numerical values come from the poles.
   - The numerator coefficients are called residues.

3. Determine the residues

   - Plug-and-chug method
   - Cover-up method
   - Graphical method
   - Matlab `residue` command

- Example, Determining the coefficients for $H(s)$ (Eqn (51)) using Matlab:

```
>> B = [ 35 82 48]          %%  Numerator polynomial
>> A = [18 47 66 40 ]       %%  Denominator polynomial

>> [Residues, Poles, NonProperPolynomial] = residue(B, A)

Residues =   0.9705 - 0.3912i
             0.9705 + 0.3912i
             0.0035

Poles =   -0.7500 + 1.1990i
          -0.7500 - 1.1990i
          -1.1111

NonProperPolynomial =    []
```

  - Using the residues found with the Matlab `residue` command, $H(s)$ is given in partial-fraction-expansion (PFE) form:

$$H(s) = \frac{0.0035}{s+1.11} + \frac{0.97-0.39\,j}{s+0.75-1.20\,j} + \frac{0.97+0.39\,j}{s+0.75+1.20\,j} \tag{55}$$

  - Eqn (55) is in the form of Eqn (52), with p_2, p_3, C_2 and C_3 all complex numbers.

- In general, if p_i is complex:

  - There will be another pole, $p_j = p_i^*$ which forms the complex conjugate pole pair.

  - The associated C_i will be complex, and will also have a complex-conjugate partner, with $C_j = C_i^*$.

  - We can also write $H(s)$ with a second order term either by cross-multiplying the two complex terms in (55), or by solving for coefficients using the plug-and-chug method. The expansion is:

$$H(s) = \frac{0.0035}{s+1.11} + \frac{1.94\,s+2.39}{s^2+1.5\,s+2} \tag{56}$$

8.3 Product-of-terms form (or root-locus, series-realization or cascade form)

- For residue calculation or the Root Locus design method, it is handy to put the TF in product-of-terms form (Note K_{rlg}, the 'root locus gain').

$$H(s) = K_{rlg} \frac{\text{Product of zero terms}}{\text{Product of pole terms}} \tag{57}$$

- The product-of-terms form also handy because it directly shows the zeros and poles.
- To create the product-of-terms form:
  1. Pull out the root locus gain, K_{rlg} (this will make each of the *highest* order coefficients 1.0):

$$H(s) = \frac{35}{18} \frac{s^2 + 2.34\,s + 1.37}{s^3 + 2.61\,s^2 + 3.67\,s + 2.22}$$

     - Find the roots of the remaining polynomials. With the z_i and p_i known, Eqn (58) can be written down directly.
     - If there are complex zero pairs and/or complex pole pairs, they can be combined to make 2nd order terms, as shown in Eqn (59).

  - A high-order filter with TF $H(s)$ can be built by building a low-order filter for each term, and combining them series.

8.3.1 General product-of-terms form:

$$H(s) = K_{rlg} \frac{(s - z_1)(s - z_2)\cdots}{(s - p_1)(s - p_2)\cdots} \quad , \qquad \text{or} \tag{58}$$

$$= K_{rlg} \frac{(s - z_1)(s - z_2)\cdots}{(s - p_1)\left(s^2 + a_1 s + a_2\right)(s - p_4)\cdots} \tag{59}$$

8.3.2 Product-of-terms form example:

The example transfer function gives:

$$H(s) = \frac{35}{18} \frac{(s+1.2)(s+1.4)}{(s+1.11)(s+0.75-1.20\,j)(s+0.75+1.20\,j)} \tag{60}$$

Or

$$\begin{aligned} H(s) &= \frac{35}{18} \frac{(s+1.2)(s+1.4)}{(s+1.11)(s^2+1.5\,s+2.0)} \\ &= \frac{35}{18} \quad \frac{(s+1.2)}{(s+1.11)} \quad \frac{(s+1.4)}{(s^2+1.5\,s+2.0)} \end{aligned} \tag{61}$$

Where the final form, Eqn (61), is suitable for implementation as two filters in series.

8.4 Summary on forms

- We've seen 3 forms for transfer functions

  1. Rational polynomial form (which can be root-locus form, or Bode form)
  2. Sum-of-terms form
  3. Product-of-terms form

- If we are implementing a filter, the three forms lead to three different filter realizations, seen in figures 12-14.

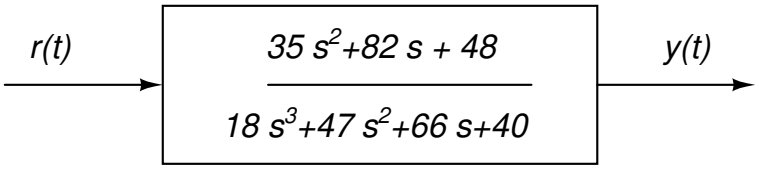


Figure 12: Rational polynomial form.

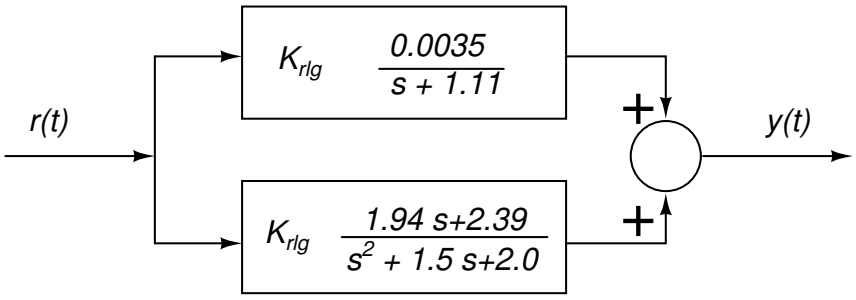


Figure 13: Sum-of-terms form.

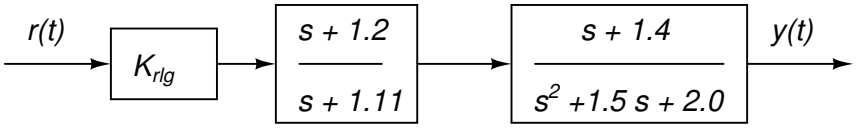


Figure 14: Product-of-terms form.

9 Transfer function and simplifying block diagrams (Franklin *et al.* sec 3.2 (7th ed.))

- The block diagram is a graphical language for representing
  - Mathematical modelsof devices,
  - The interconnectionsof devices.
- It is the graphical language of the Controls Engineer[1].
- Sometimes, a system can give a rather (or even very) complicated block diagram;
  - It is necessary to reduce the block diagram to a single transfer function.
- Consider figure 15. The system has 3 loops, two feed-back and one feed-forward.

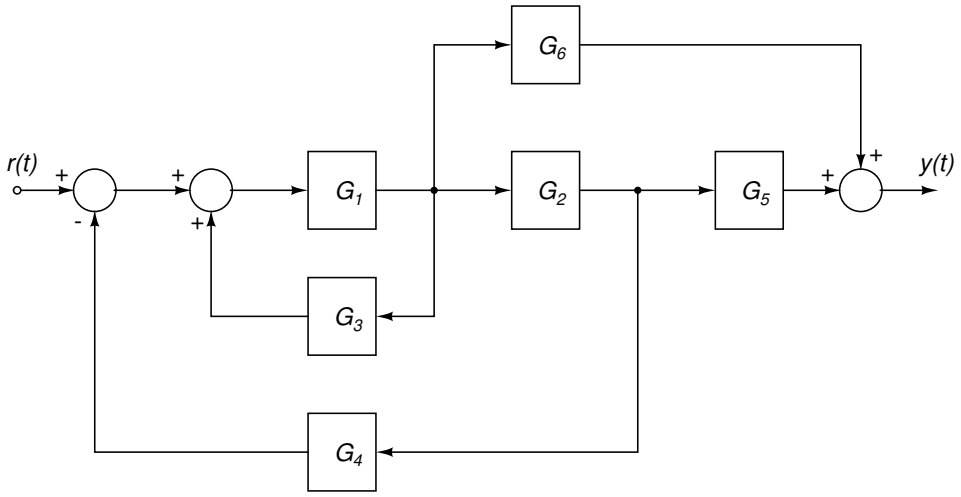


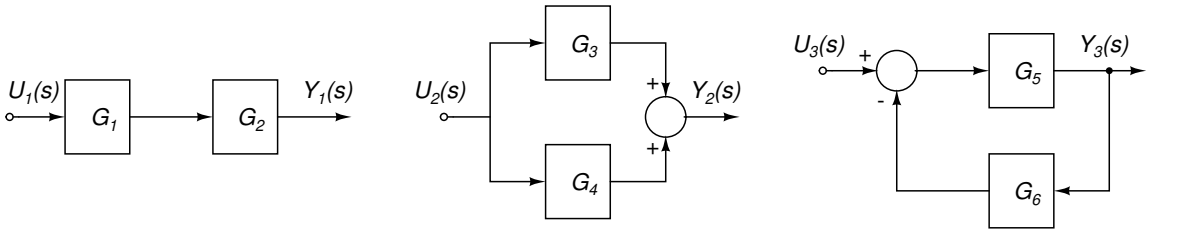
Figure 15: Complicated block diagram.

- We need to be able to determine $G_{uy}(s)$ for possibly complicated block diagrams.

1. For example, Simulink is Mathwork's Matlab tool for directly, graphically programming a numerical simulation in with block diagrams. (No wonder Mathwork's drove out the competition !).

9.1 Basic block diagram algebra

- First we'll review some basic algebra of block diagrams.
- Recall that block diagrams represent mathematical relationships.
  - Each transformation of a block diagram corresponds to an algebraic step.
  - Every algebraic step has a corresponding block-diagram transformation.
- Figure 16 shows the most basic block diagram transformations.



Series: $\frac{Y_1(s)}{U_1(s)} = G_1 G_2$ Parallel: $\frac{Y_2(s)}{U_2(s)} = G_3 + G_4$ Feedback: $\frac{Y_3(s)}{U_3(s)} = \frac{G_5}{1+G_5 G_6}$

Figure 16: Basic block diagram algebra.

- Each diagram of figure 16 can be transformed to the block of figure 17, with the algebra as given in figure 16.

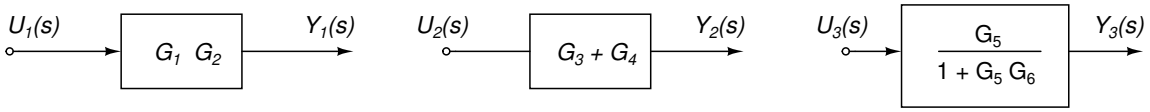
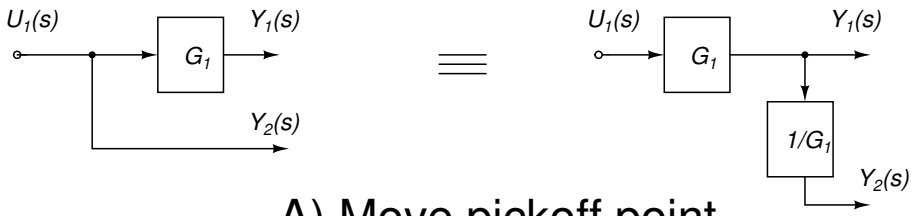
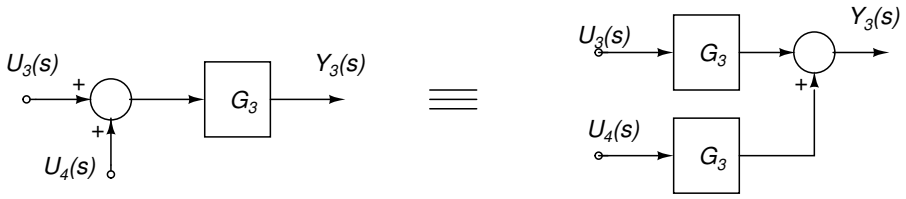


Figure 17: Equivalent block diagram.

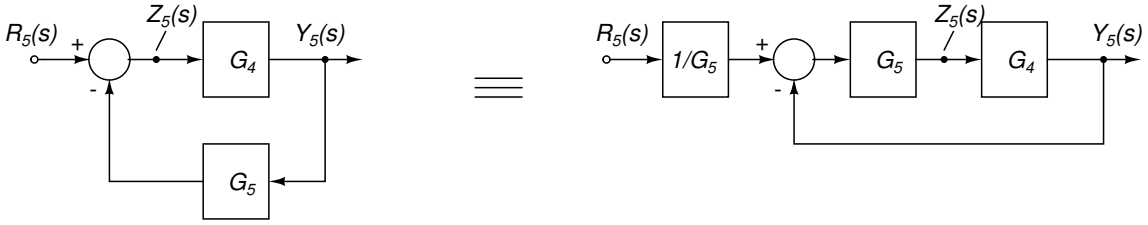
More basic block diagram algebra



A) Move pickoff point



B) Move summing junction



C) Move gain block through summing junction

Figure 18: Additional examples of block diagram algebra.

- Figure 18 illustrates additional examples of block diagram algebra.
- The blocks implement these relationships:

$$(A) \quad Y_1 = G_1 U_1; \quad Y_2 = U_1 \implies Y_1 = G_1 U_1; \quad Y_2 = G_1 \frac{1}{G_1} U_1$$

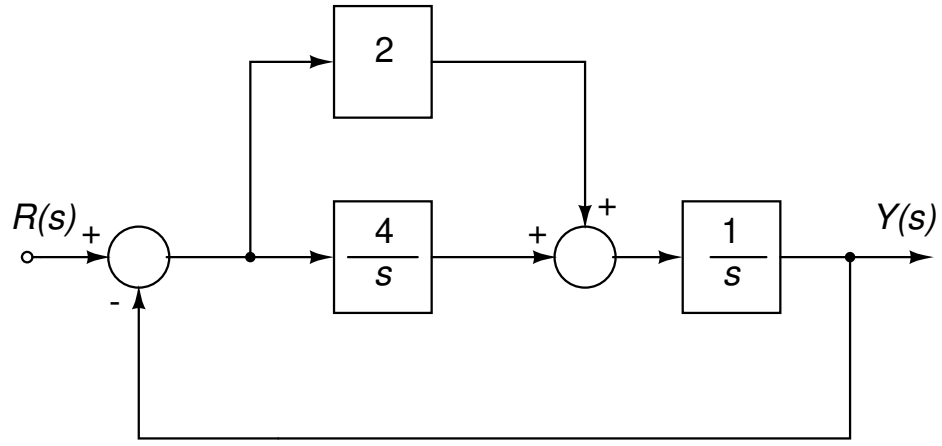
$$(B) \quad Y_3 = G_3 (U_3 + U_4) \implies Y_3 = G_3 U_3 + G_3 U_4$$

$$(C) \quad Y_5 = \frac{G_4}{1 + G_4 G_5} R_5 \implies Y_5 = \frac{1}{G_5} \frac{G_4 G_5}{1 + G_4 G_5} R_5$$

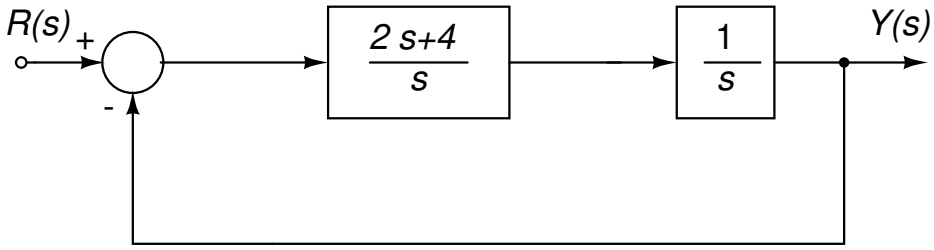
- These examples also make the point that block diagrams are a graphical language for expressing functional relationships.

9.1.1 Example

- An example of block diagram reduction is seen in figure 19.



A) Original diagram



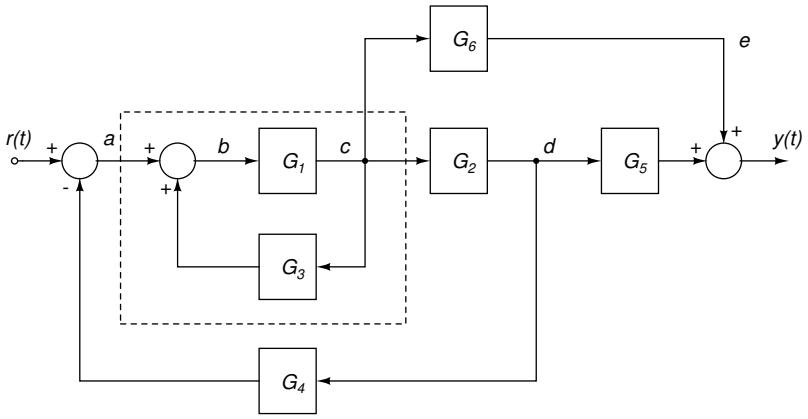
B) Simplified diagram

Figure 19: Example of block diagram reduction.

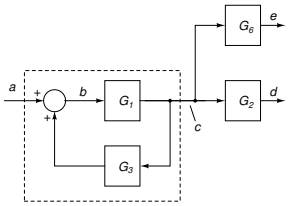
  - Combine parallel blocks of (2) and $(4/s)$.
  - Put $G_1(s) = (2+4/s)$ into rational polynomial form.

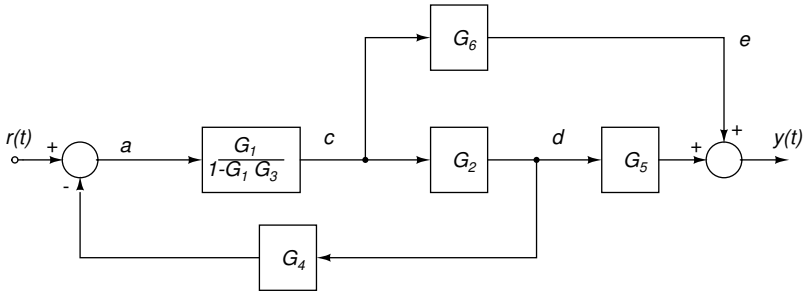
- What is the transfer function $T_{ry}(s)$? Student Exercise

9.1.2 Second example

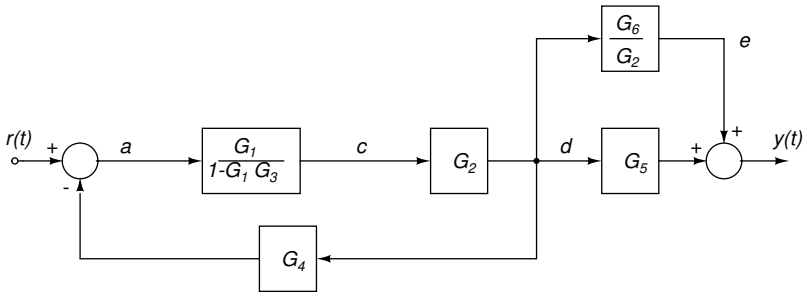


A) Original diagram. Note: we can move the pick-off point for G_6 to the right.

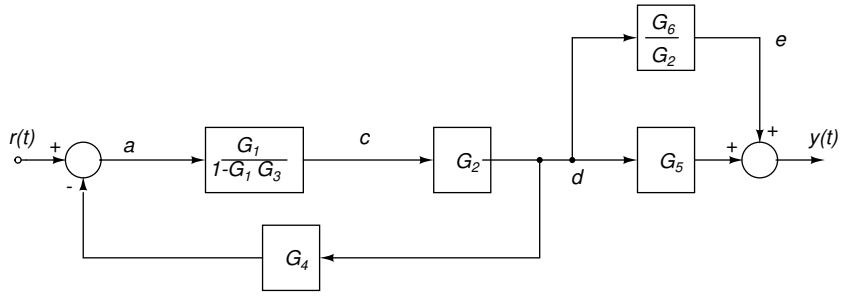




B) Simplify inner feedback loop. This leaves two coupled loops.



C) Pass the pick-off point through G_5 to decouple feedback and feed-forward loops.



D) Illustrate that the two loops are truly decoupled.

Figure 20: Additional example of block diagram reduction. Signals are marked with letters *a* ... *e*, to show equivalent signals across figures.

- The block diagram of figure 20 can be reduced to 21. Note that of the original signals, b, c and e are gone from the model. Block diagram simplification removes signals as well as blocks from the model.

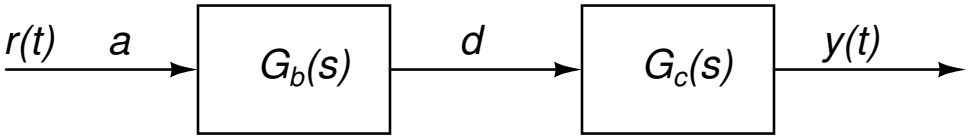


Figure 21: Reduction of block diagram in figure 20.

1. What are transfer functions $G_b(s)$ and $G_c(s)$? Student Exercises

2. What is the transfer function $T_{ry}(s)$?

9.1.3 Summary, block diagram reduction

- Doing algebra on a block diagram combines or eliminates signals, just as algebra on equations combines or eliminates variables.
  - Reducing a block diagram normally involves removing physical signals.
  - Reducing a block diagram can involve introducing non-physical signals.
  - The input and output signals must remain unchanged, because they connect to the outside world.

10 Similarities and Differences between an LT and a TF

- Laplace transforms and transfer functions can look quite similar:
  - Laplace Transform: if $y(t) = 3.25\,e^{-3t}\cos(4t+22^o)$ then $\mathcal{L}\{y(t)\}$, and the **LT** will be:
$$Y(s) = \frac{3s+4}{s^2+6s+25} \tag{62}$$
  - Transfer Function: if $\ddot{y}(t)+6\dot{y}(t)+25\,y(t) = 3\,\dot{u}(t)+4\,u(t)$ then
$$Y(s)\,\left(s^2+6s+25\right) = U(s)\,(3s+4)$$
And the **TF** will be:
$$G_p(s) = \frac{3s+4}{s^2+6s+25} \tag{63}$$
- But the meanings of the LT and TF are quite different:

	Laplace Transform	Transfer Function
Derived from a system description		√
Derived from a time domain signal.	√	
It has an inverse.	√	
It has units.	√	√
It is a rational polynomial.	√	√
Formed as a ratio of Laplace Transforms		√
Generally known to the controls engineer		√

Similarities Differences between an LT and a TF (continued)

- We determine from context whether a rational polynomial refers to:

  1. An LT, which is to say a signal or function of time;

  2. A TF, which is to say the dynamics of a component or system.

- The LT and TF are connected through the impulse response:

  If $u(t)$ is an impulse,

  $$U(s) = 1$$

  then

  $$Y(s) = G_p(s)\, U(s) = G_p(s)$$

  .

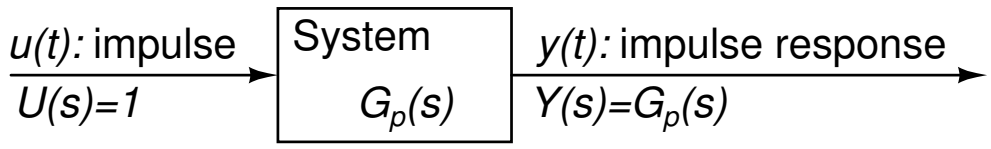


Figure 22: The Laplace Transform of the output is the Transfer Function, if the input is an impulse.

11 Conclusions

In this part, we've looked in more detail at transfer functions. We've seen

- A general notion of a system, as an element that processes an input signal to produce an output signal
- We've looked again at forming the transfer function, and seen a detailed example exploring the units of the terms in a transfer function
- Algebraic manipulations that are allowed with transfer functions
- Transfer functions and block diagrams
  - How transfer functions are formed from differential equations
  - How block diagrams are simplified